

MUHAMMAD YAFIZHAM BATUBARA

AI Engineer | Backend, Agentic AI & LLM Systems

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PROFESSIONAL SUMMARY

AI Engineer specializing in production AI systems and enterprise backend engineering. Building agentic workflows, RAG pipelines, and AI microservices with Python, FastAPI, and LangGraph, backed by a proven track record of 14+ production business systems built and operated as a sole engineer. Self-taught developer with a strong engineering background; fluent in English (IELTS Band 7).

TECHNICAL SKILLS

- Agentic AI & LLM Systems:** Python, LangGraph, LangChain, RAG Pipelines, Vector Databases (Pgvector, Qdrant), Async Systems, Structured Outputs.
- LLMOps & Observability:** Langfuse Tracing, Token/Cost Monitoring, Latency Optimization, Dataset Evaluations, Regression Testing.
- Backend & Integrations:** FastAPI, Pydantic, TypeScript, NestJS, Node.js, Express.js, React, Next.js, RESTful APIs, Redis.
- Databases & Infrastructure:** PostgreSQL, MongoDB, Docker, Linux VPS, Nginx, CI/CD Pipelines (GitHub Actions).

WORK EXPERIENCE

PT Catur Sukses Internasional — Software Engineer

October 2024 – Present (Indonesia)

- System Architecture:** Sole engineer responsible for architecting, building, deploying, and maintaining 14+ production enterprise applications across logistics, sales, finance, and workflow automation.
- Performance Optimization:** Integrated Redis caching and optimized transactional PostgreSQL queries, reducing database load and resolving conflict detection in real-time warehouse operations.
- Business Translation:** Translated complex spreadsheet-based operational workflows into highly scalable production code, eliminating manual operations for internal stakeholders.

KEY ENGINEERING PROJECTS

AI-Powered Workflow & Approval Builder | *Node.js, Express, MongoDB, React, Active Directory* — [Source Code](#)

- System Architecture & AD/LDAP:** Integrated Active Directory (LDAP) authentication to dynamically map user accounts, departments, and roles directly from enterprise directory servers.
- Multi-Tenant Operations:** Engineered a cross-organization flow duplication engine, allowing super-admins to instantly replicate complex multi-step approval workflows across independent tenant domains.
- Granular Permissions & Notifications:** Designed a department-scoped permission matrix with automated real-time email dispatch for workflow state transitions (approvals, rejections, takeovers, and rollbacks).

Real-Time Multi-Warehouse Management Platform | *NestJS, PostgreSQL, Redis, TypeScript, Next.js* — [Source Code](#)

- Conflict Resolution & Rescheduling Algorithm:** Built an intelligent scheduling algorithm that automatically resolves dock booking collisions by dynamically calculating and suggesting the next available open slots.
- Real-Time Live Queue Management:** Designed a real-time live queue interface with complex drag-and-drop workflows (unloading state transitions) powered by NestJS and Redis cache locks.
- Data Integrity:** Enforced strict relational constraints and query optimizations in PostgreSQL, deliberately replacing legacy database Enums with flexible lookup models to accommodate evolving business rules.

Enterprise AI Workflow Assistant | *Python, FastAPI, LangGraph, NestJS, Pgvector, Docker* — *In Development*

- Building a multi-agent workflow system using Python, FastAPI, and LangGraph to automate repetitive marketing operations, including partner research, content generation, and customer service workflows.
- Designing RAG pipelines with PostgreSQL and Pgvector to provide context-aware retrieval over business knowledge and operational data.
- Implementing LLMOps & Observability via Langfuse to trace agent execution paths, monitor token cost/latency, and run dataset-based evaluations.
- Exposing AI agent workflows via async FastAPI microservices, seamlessly integrated with NestJS enterprise backend services.

Enterprise POS & Inventory Platform | *JavaScript, Node.js, MongoDB, React Native* — [Source Code](#)

- Fraud & Double-Spending Prevention:** Engineered a distributed voucher redemption service with real-time online validation and ledger locks, preventing double-spending of retail vouchers across separate offline-first physical outlets.
- Cross-Platform Verification (Web-Mobile):** Built a secure, real-time transaction voiding protocol where void requests initiated via React Native mobile POS must be confirmed and authorized via the NestJS Web admin panel.
- Mobile Cache Optimization:** Optimized React Native client performance by scoping AsyncStorage synchronization queries strictly to the cashier's active outlet, reducing memory footprint and startup latency.
- Hardware & Printing Integration:** Programmed print buffers and custom footer layout configurations (tax/NPWP, localized branch addresses) directly integrating with thermal receipt printers.

EDUCATION

Undergraduate Coursework in Mining Engineering

Institut Sains dan Teknologi Medan — Completed 6 Semesters (3 Years)

- Pivoted to software engineering through intensive self-directed study; completed coursework in Advanced Engineering Mathematics, Statistics, and Physics.
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LANGUAGES & CERTIFICATIONS

- **English:** IELTS General Training – Band 7 (Speaking: 7, Listening: 7, Reading: 7, Writing: 6)